

TrES-3b

R-2750

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190704 · created 2026-09-02T23:18:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458668.7449 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.034
Transit depth (Rp/Rs)^2	3.61 +/- 1.31 [%]
Orbital Inclination (inc)	81.13 +/- 1.5
Airmass coefficient 1 (a1)	1.359 +/- 0.014
Airmass coefficient 2 (a2)	-0.296 +/- 0.035
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	4.01
Optimal Annulus	13.79
Transit Duration (day)	0.046 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.034 vs 0.16309 (deviation 0.78σ)
Duration fit vs published	0.046 d vs 0.0567 d (deviation 0.52σ)
Mid-time O–C	2.7 min
Depth SNR	5.5
Residual scatter	1.05 %

Light curve

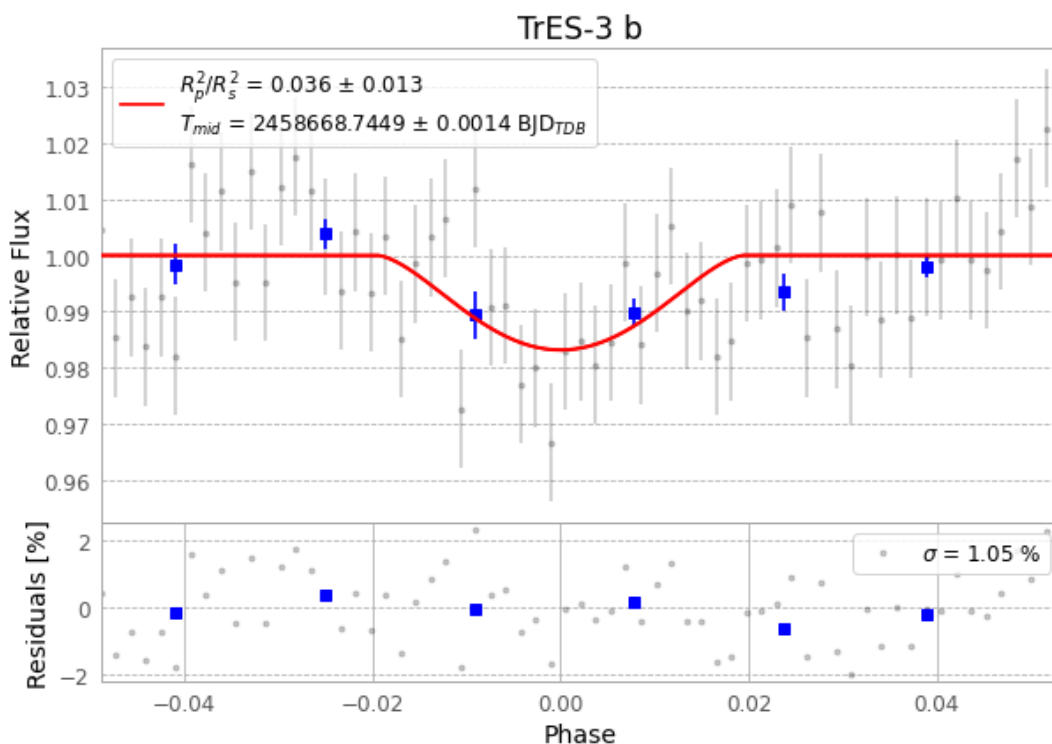


Figure 1 — FinalLightCurve_TrES-3 b_2019-07-03.png (SHA-256 3c71a3c2afea1c56...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 141], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 24289abd41180081...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3190704041512.FITS → TRES-3190704072712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190704.log (75cb90c06baf...), FinalLightCurve_TrES-3 b_2019-07-03.png (3c71a3c2afea...), FinalLightCurve_TrES-3 b_2019-07-03.csv (b66c7db13e7d...)

Compute resources

Wall time 162.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 23:18Z.